[OPTIONAL] Prometheus Metrics Reporting with Grafana Visualization

A metrics-native alternative to three of the seven Flink reports: emit them as Micrometer meters from the producer interceptor and let Prometheus do the windowing at read time. **Additive and opt-in** — it does not replace the Flink reports. The companion one-command showcase (Prometheus + Grafana on minikube) has its own runbook: k8s/monitoring/README.md.

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1.0 Why three of seven

Is Flink overkill for these reports? **Yes and no** — it's not all about stateless aggregation. Of the seven reports, **three are pure stateless scalar aggregation** keyed on bounded-cardinality dimensions (service / topic / hop_count — never `trace_id`):

- **`latency_1m`** (10_latency_report.fql) — avg/max latency per `(pipeline, origin_service, this_topic)`.
- **`topology_1m`** (20_topology_report.fql) — produce-edge record counts.
- **`hop_distribution_1m`** (30_hop_distribution.fql) — records per `hop_count`.

These don't need a stream processor. The producer interceptor already has every value in scope on each `send()`, so it can emit them as **Micrometer** meters and let **Prometheus** do the 1-minute windowing at query time (`rate()` / `increase()`), with **Grafana** on top. The other four reports — `latency_percentiles` (mergeable T-Digest), `coverage`, `bipartite_topology`, `stuck_trace` — are per-`trace_id` *stateful* or *absence-of-event* problems Prometheus can't express, so they **stay in Flink**. This path is **additive and opt-in** (off by default); it doesn't replace the Flink reports — it's the cheaper way to serve the three that are metrics, not stream processing.

Emission lives in one place — PrometheusIsotopeMetrics.java — mirroring how `TDigests` centralizes the sketch contract.

> **Packaged as a library.** The tracing is an adoptable, publishable library, split so metrics are optional: `kafka-isotope-core` carries the propagation (interceptor, headers, consume markers — Jackson + Kafka only), and `kafka-isotope-metrics` adds the Micrometer/Prometheus exporter. The core routes emissions through a no-op `IsotopeMetricsSink` until `kafka-isotope-metrics` registers the Prometheus one, so propagation pulls no metrics dependency. The runnable stages below (`:app`) are the reference consumer — see kafka-isotope-core/README.md to adopt it elsewhere.

2.0 Enabling the exporter

Pass `-Dmetrics.prometheus.enabled=true` to any long-running stage. **Producing** stages (`hop` / `enrich` / `fulfill`) emit the produce-side meters via the interceptor; **consuming** stages emit the consume-side meters via the marker path (Consume-side meters) — `hop` does both (it consumes *and* produces), and the terminal `consume` / `ship` stages emit the consume side alone. The app serves the meters at `GET /metrics` on `metrics.prometheus.port` (default `9404`) via the JDK's built-in HTTP server — no extra runtime.

```
# Prereq: 'make kafka-pf-up' is up. Run the enrich stage with metrics
exposed:
./gradlew :app:run -Dmetrics.prometheus.enabled=true --args="enrich"
# → metrics on http://localhost:9404/metrics

# In another shell, drive traffic so the meters move (origin produce):
for i in {1..30}; do ./gradlew :app:run --args="place burst-$i" -q; sleep 5;
done

# Scrape:
curl -s localhost:9404/metrics | grep isotope_
```

Point a Prometheus scrape job at each stage's `/metrics` and the three reports become PromQL queries (no report topics, no Control Center — Grafana instead). Unlike the Flink jobs, there's **no watermark wait**: cumulative counters update on every send and Prometheus windows them at read time.

> **Running the Grafana showcase? Add `-Dmetrics.prometheus.port`.** The command above binds the default `9404`, which is right for a single stage you scrape by hand — but stages share one host, so they cannot all bind it, and §8.0's Prometheus config scrapes a fixed port per stage: `enrich`→`9410`, `fulfill`→`9411`, `ship`→`9412`. Omit the flag and the stage still exports happily on `9404` while every Prometheus target reads `connection refused` and Grafana shows no data. Check with `kubectl port-forward -n monitoring svc/prometheus 9090:9090` → Status → Targets.

> **Reading the latency metric — mind the backlog.** `hop` / `consume` use a *random* consumer group with `auto.offset.reset=earliest`, so a fresh stage **replays the whole topic from offset 0**. Latency is `now − origin_ts`, so days-old replayed records report enormous values — e.g. an `isotope_hop_latency_seconds_sum` of ~9,000,000 over 58 records (~43 h average) is just the backlog, not steady-state latency (the Flink `latency_1m` report shows the same, by the same definition). The tell: `_count` exceeds the number of records you just sent. Two ways to see realistic latency: drain the backlog and read `rate(isotope_hop_latency_seconds_sum[1m]) / rate(..._count[1m])` (the windowed rate ignores the stale backlog once it stops growing), or skip

> the backlog entirely with `-Disotope.consume.from=latest` so the stage only consumes records produced after it starts:
>
> ```
> ./gradlew :app:run -Dmetrics.prometheus.enabled=true -
> Disotope.consume.from=latest --args="enrich"
> ```

3.0 The three reports as PromQL

Two meters cover all three reports — a `Timer` whose count doubles as the topology edge count, plus a `Counter` for the hop histogram:

Report	Meter (Prometheus name)	Labels
`latency_1m` + `topology_1m`	`isotope_hop_latency_seconds_{count,sum,max}` (`Timer`)	`pipeline`, `origin_service`, `this_service`, `this_topic`
`hop_distribution_1m`	`isotope_hop_records_total` (`Counter`)	`pipeline`, `this_topic`, `hop_count`

```
# latency_1m — avg latency (seconds)
  sum by (pipeline, origin_service, this_topic)
(rate(isotope_hop_latency_seconds_sum[1m]))
/ sum by (pipeline, origin_service, this_topic)
(rate(isotope_hop_latency_seconds_count[1m]))

# latency_1m — max
max by (pipeline, origin_service, this_topic)
(isotope_hop_latency_seconds_max)

# topology_1m — records per produce edge
sum by (pipeline, origin_service, this_service, this_topic)
(increase(isotope_hop_latency_seconds_count[1m]))

# hop_distribution_1m — records per hop_count
sum by (pipeline, this_topic, hop_count)
(increase(isotope_hop_records_total[1m]))
```

4.0 Consume-side meters

The three meters above all come from the **produce** side — the interceptor on `send()`. The **consume** side adds three more. Two — the edge counter and the time-to-consume latency — are emitted by `IsotopeContext.recordConsume` right beside the consume-edge marker it writes to `isotope_consume_edge_markers`. The third, `isotope.consume.age`, is emitted **once per consumed record on whichever path the consumer takes**: continuing consumers report it from the **adoption path** (`IsotopeContext.adoptFromRecord(record, service)`), and terminal consumers — which never

adopt — report it from the **marker path** (`recordConsume`, guarded on `current() == null` so a stage that does both, like `hop`, never double-counts). So age fires on every consuming stage: `hop` via adoption, the terminal `consume` / `ship` stages via the marker. Same gating — no-op unless the exporter is started.

Signal	Meter (Prometheus name)	Labels
consume-edge counts (topic→consumer)	`isotope_consume_records_total` (`Counter`)	`pipeline`, `this_topic`, `consumer_service`
time-to-consume latency (origin→consume)	`isotope_consume_latency_seconds_{count,sum,max}` (`Timer`)	`pipeline`, `origin_service`, `consumer_service`, `this_topic`
origin→adoption age (how stale at pickup)	`isotope_consume_age_seconds_{count,sum,max}` (`Timer`)	`pipeline`, `origin_service`, `consumer_service`, `this_topic`

```
# consume edges — records consumed per (topic → consumer) per minute
sum by (pipeline, this_topic, consumer_service)
(increase(isotope_consume_records_total[1m]))

# time-to-consume — avg seconds from trace origin to consumption
  sum by (pipeline, consumer_service, this_topic)
(rate(isotope_consume_latency_seconds_sum[1m]))
/ sum by (pipeline, consumer_service, this_topic)
(rate(isotope_consume_latency_seconds_count[1m]))

# consume age — avg seconds a record had aged by the time a service adopted
it
  sum by (pipeline, consumer_service, this_topic)
(rate(isotope_consume_age_seconds_sum[1m]))
/ sum by (pipeline, consumer_service, this_topic)
(rate(isotope_consume_age_seconds_count[1m]))
```

These are **net-new signals**, not a port of a Flink report. `isotope_consume_records_total` is the *consume half* of the bipartite topology — the topic→consumer edge tallies — and `isotope_consume_latency_seconds_*` is an origin→consume figure no Flink report computes (the Flink `latency_1m` measures origin→**produce**-hop). The **full** `bipartite_topology` report still stays in Flink, but not because it follows individual traces: it's a `UNION ALL` of per-minute produce-edge and consume-edge tallies, and those edge counts are exactly what these meters reproduce. What a counter can't reproduce is its `distinct_traces` column — `COUNT(DISTINCT trace_id)` per edge per window (see What stays in Flink). What you get here is the bounded-cardinality edge *counts* and a time-to-consume distribution — cheap, windowable at read time, and free of the watermark wait.

`isotope_consume_age_seconds_*` measures the **same `now - origin_ts` quantity** as `isotope_consume_latency_seconds_*`, but it's the **universal consume-side age signal** — emitted once on *every* consuming stage, where latency only fires on the marker path. That's the point of having

both: query `isotope_consume_age_*` to get consume-lag across **all** consumers (including `hop`-style stages that adopt-and-forward without ever writing a marker), and the timer's `_count` doubles as the per-`(consumer_service, topic)` consume rate. At a terminal consumer the age equals that consumer's latency by definition (same `now - origin_ts`); the two diverge in *coverage*, not value — latency is marker-only, age is everywhere.

> The same **backlog caveat** from Enabling the exporter applies: `hop` / `consume` / `ship` use a random consumer group with `auto.offset.reset=earliest`, so a fresh stage replays from offset 0 and both `isotope_consume_latency_seconds_sum` and `isotope_consume_age_seconds_sum` then reflect days-old origins, not steady-state. Read the windowed `rate(...sum[1m]) / rate(...count[1m])`, or start with `-Disotope.consume.from=latest`.

5.0 Operational queries (cookbook)

The queries above mirror the three reports. These are the **operational** angles you'd reach for when running the demo or alerting on it — they lean on the `status` label (present on every meter: `success` on the happy path) and on read-time ratios. All are `pipeline`-scoped via `{pipeline=~"$pipeline"}` when used in the provisioned dashboard.

```
# Hop failure ratio per service (0–1) — clamp the denominator so a quiet
# stage reads 0, not NaN. Grafana: format as "percent (0.0–1.0)".
  sum by (this_service) (rate(isotope_hop_records_total{status!="success"}
[5m]))
/ clamp_min(sum by (this_service) (rate(isotope_hop_records_total[5m])), 1e-
9)

# Hop completion — share of traffic reaching the deepest hop (full pipeline).
# hop_count="3" is the terminal depth for the orders.* chain.
  sum(rate(isotope_hop_records_total{hop_count="3"}[5m]))
/ clamp_min(sum(rate(isotope_hop_records_total[5m])), 1e-9)

# Consumers reading data older than 30s on average — adoption-lag alert.
( sum by (consumer_service) (rate(isotope_consume_age_seconds_sum[5m]))
/ sum by (consumer_service) (rate(isotope_consume_age_seconds_count[5m])) ) >
30

# Stage liveness — fires for any stage that emitted no record in the last 2m.
sum by (this_service) (increase(isotope_hop_records_total[2m])) == 0

# Scrape health — an exporter is down (host stage not running on its port).
up{job="isotope-stages"} == 0

# Drop-off between adjacent topics — records into enriched vs into fulfilled.
  sum(rate(isotope_consume_records_total{this_topic="orders.enriched"}[5m]))
- sum(rate(isotope_consume_records_total{this_topic="orders.fulfilled"}[5m]))
```

> **Why no `histogram_quantile`?** These are Micrometer `Timer`s exported as `_sum`/`_count`/`_max` — there are **no `_bucket` series**, so percentiles aren't available. Use `rate(_sum)/rate(_count)` for the average and `_max` for the recent worst case (a decaying per-step max, not an all-time high). Enable client-side histograms in the exporter if you need true quantiles.

6.0 Mapping questions to PromQL

What can you actually *ask* these meters? Below, each question carries a verdict:

- ✅ **PromQL** — answerable directly from the meters.
- 🟡 **PromQL (approx)** — answerable as a bounded-cardinality *aggregate*; the exact, per-trace form stays in Flink (or the record header).
- 🔴 **Header / Flink** — per-`trace_id` or absence-of-event; **no Prometheus equivalent** (these meters never tag `trace_id` — see PrometheusIsotopeMetrics.java).

6.1 Easy — single record, single trace

Every question here is about **one** record or trace. Prometheus aggregates away identity, so the answer lives on the record's `isotope` header (inspect the record) or in Flink's per-trace reports — **not** in these meters.

Question	Verdict	Where to look
Did my record get tagged?	🔴	The `isotope` header on the record itself — presence = tagged.
What's the origin of this trace?	🔴	`origin_service` / `origin_ts` fields in the header.
How many hops has *this* record taken?	🔴	The header's hop list length. (The *distribution* across all records is ✅ — see Easy→Medium.)
Where did this trace go?	🔴	Replay the record's `isotope` header hop list (consume side: filter `isotope_consume_edge_markers` by `trace_id`). Flink `bipartite_topology` aggregates edges per window — it doesn't keep per-trace paths.
Did the trace ID survive a consume-then-produce hop?	🔴	App/Flink per-trace; a counter can't follow one `trace_id`.
Which pipeline does this trace belong to?	🔴	Header `pipeline` field. (`pipeline` is a meter *label*, but can't isolate one trace.)

6.2 Easy → Medium — single per-minute aggregates

This is the meters' sweet spot: bounded-cardinality scalar aggregation, windowed at query time.

```
# ✅ End-to-end latency intake → shipping over the last minute
(origin→consume)
  sum(rate(isotope_consume_latency_seconds_sum{consumer_service="shipping-
notification-service"}[1m]))
/ sum(rate(isotope_consume_latency_seconds_count{consumer_service="shipping-
notification-service"}[1m]))
```

```
# 🟡 The actual service graph — produce edges + consume edges (the edge
*sets*;
#    Flink's bipartite_topology adds distinct_traces per edge)
sum by (this_service, this_topic)
(increase(isotope_hop_latency_seconds_count[1m]))   # produce side
sum by (this_topic, consumer_service)
(increase(isotope_consume_records_total[1m]))       # consume side

# 🟡 Records per topic per minute — proxy for "distinct traces"; a counter
can't
#    COUNT(DISTINCT trace_id), so this counts records, not deduped traces (→
Flink)
sum by (this_topic) (increase(isotope_hop_records_total[1m]))

# ✅ Hop-count distribution — long tails here = retry storms
sum by (hop_count) (increase(isotope_hop_records_total[1m]))

# ✅ Traces hitting the 32-hop ceiling (Isotope.MAX_HOPS). The eviction
*marker*
#    itself is an absence-of-event → Flink; the count at the ceiling is
metric-native
sum (increase(isotope_hop_records_total{hop_count="32"}[5m]))

# 🟡 Records each pipeline carries, minute by minute (records, not distinct
traces)
sum by (pipeline) (increase(isotope_hop_records_total[1m]))
```

6.3 Medium — cross-window deltas, anomalies, multi-report joins

Cross-window deltas and "newly appeared" detection are a Prometheus *strength*; per-trace coverage and stuck-trace detection are where it hands off to Flink.

```
# ✅ Did latency get worse after the 2pm deploy? — now vs. before, via
offset
  ( sum(rate(isotope_hop_latency_seconds_sum[5m])) /
sum(rate(isotope_hop_latency_seconds_count[5m])) )
- ( sum(rate(isotope_hop_latency_seconds_sum[5m] offset 2h)) /
sum(rate(isotope_hop_latency_seconds_count[5m] offset 2h)) )

# 🟡 What % of intake records made it through to the shipping consumer?
#    Ratio of rates — NOT a per-trace coverage join (that's Flink's coverage
report)
  sum(rate(isotope_consume_records_total{consumer_service="shipping-
notification-service"}[5m]))
/ sum(rate(isotope_hop_records_total{this_topic="orders.placed"}[5m]))

# 🟡 Where are records being dropped? — drop-off between adjacent edges.
#    Pinpointing *which* traces dropped = Flink coverage
  sum(rate(isotope_hop_records_total{this_topic="orders.enriched"}[5m]))
- sum(rate(isotope_hop_records_total{this_topic="orders.fulfilled"}[5m]))

# ✅ When a new service goes live, when does it first appear in the
topology?
```

```
#    Series present now but absent an hour ago
  group by (this_service) (isotope_hop_latency_seconds_count)
unless
  group by (this_service) (isotope_hop_latency_seconds_count offset 1h)
```

Question	Verdict	Why
Are some traces duplicated at the terminal sink?	🔴	Dedup is per-`trace_id`; a counter can hint (consume > produce) but can't name the dupes → Flink.
Which traces went in but never came out within 60s?	🔴	Absence-of-event per trace = Flink `stuck_trace_alerts`; Prometheus can't assert a *missing* series.
Where did each stuck trace last get seen?	🔴	Per-trace state → Flink / the header trail.

The pattern: **aggregates and time-deltas → PromQL; identity, dedup, and absence → Flink.** That boundary is the same one PrometheusIsotopeMetrics.java draws (3 metrics-native reports, 4 stay in Flink), and the next section spells out the two columns that can never cross it.

6.4 Hard — tail latency, drift, correlation

Past this point PromQL is out of the running — not because the questions are exotic, but because every one of them needs either an adaptive sketch or a join across two reports. Listed to complete the five-tier ladder the root README shows; the "where to look" column is the answer.

Question	Verdict	Where to look
What are p50 / p95 / p99 across the pipeline?	🔴	`latency_percentiles_flat_1m` — a T-Digest sketch inside `LATENCY_PERCENTILES`, keyed `(pipeline, origin_service, this_topic)`. These Timers publish no `_bucket` series, so `histogram_quantile` is unavailable (§5.0, §7.0).
Which edge owns the tail?	🔴	Same report — rank `p99_ms` across `this_topic` inside one window.
Did the tail get worse after a deploy?	🔴	Compare `p99_ms` for one key across two windows. PromQL's `offset` trick in §6.3 drifts the **average**, never the tail.
Is a tail regression global, or confined to one pipeline?	🔴	`p99_ms` per `pipeline` in the same window; `sample_count` says whether the key carried enough traffic to trust the number.
Are the slow windows the retry-storm windows?	🔴	Join `hop_distribution_1m` to `latency_percentiles_flat_1m` on `(window_start, pipeline, this_topic)` — hop-count outliers against the tail.
Is the average hiding the tail?	🔴	`avg_latency_ms` / `max_latency_ms` from `latency_report_1m` against `p99_ms` on the same key. Prometheus ports avg and max, but

Question	Verdict	Where to look
		not the windowed min (§7.0).

6.5 Harder — forensic replay, compliance, cross-system

Identity, derivation, and absence — the three things a counter structurally cannot hold. Two of these need an opt-in stage, marked inline.

Question	Verdict	Where to look
Reconstruct a full per-trace journey for an audit?	🔴	The record's own `isotope` header hop list, plus `bipartite_topology_report_1m` for the edges around it and `stuck_trace_alerts_1m` (`last_service`, `last_topic`, `last_hop_count`) for where it stopped.
Which input records produced this merged output?	🔴	**Opt-in** fan-in provenance (flink-collector.md §2.4): join `orders.flink_batched` to `isotope_merge_edge_markers` on `merge_trace_id` for the whole parent set — `contributing_trace_id` / `_service` / `_topic` / `_records`, one row per contributing trace per window (the row count equals the merged record's `distinct_traces`; `SUM(contributing_records)` equals its `event_count`). No aggregate ever forwards a parent's trace, so this is the *only* way to recover the derivation.
Which stage lost the traces that never reached the terminal consumer?	🔴	`coverage_report_1m`'s `distinct_traces`, walked topic by topic, locates the drop-off; naming the lost traces takes an anti-join over the `isotope` view against the consume edges.
Did the isotope survive Flink's own hop?	🔴	`orders.flink_enriched` carries the hop `ISOTOPE_APPEND_HOP` stamped in-band, so Flink appears in `topology_report_1m` / `bipartite_topology_report_1m` as a `producer_service` like any service (flink-collector.md).
What fraction of records on a topic carried a parseable isotope at all?	🔴	Count `isotope_raw` (unfiltered) against the `isotope` view, which keeps only rows with `x-isotope-trace-id`. Prometheus cannot supply the denominator: an untagged record never reaches a meter in the first place.
Which versions produced the current value of this row?	🔴	**Opt-in**, CP-only state-level provenance (root README §3.5.3): `isotope_state_provenance` carries one record per emitted state, content-addressed by `StateVersion`, with the versions it derived from inline in `parents`. This is the one question the windowed merge collector cannot reach at all — an unbounded `GROUP BY`, a regular join, and a dedup have no window bounds to key an identity on.
Why was this trace stuck —	🔴	**Opt-in**, CCAF-only AI report (root README §3.3.2): `isotope_report_trace_rca_1m.root_cause`, one `ML_PREDICT` call

Question	Verdict	Where to look
not just that it was?		per stuck-trace alert.

7.0 What stays in Flink — and two deliberate gaps

Moving these out isn't free — two columns the Flink reports carry have **no Prometheus equivalent**, and both are documented in PrometheusIsotopeMetrics.java:

- **No `distinct_traces`.** All three SQL reports carry a `COUNT(DISTINCT trace_id)` column. A counter can't dedup, and `trace_id` is unbounded-cardinality so it can't be a label. If you need it, that column stays in Flink (or approximate it with a HyperLogLog sketch).
- **No windowed `min` latency.** A Micrometer `Timer` exposes max but not a per-window min — avg and max port cleanly, min does not.

So the line between "metric" and "Flink job" runs *through* a couple of these reports, not cleanly between them — which is exactly why the move is opt-in rather than a wholesale replacement.

Why `latency_percentiles_1m` is NOT in this list. Percentiles *can* be served by Prometheus — but only via a *classic* histogram (`publishPercentileHistogram()` + `histogram_quantile()`), whose accuracy is **bucket-bound, not adaptive**: error is the width of fixed, pre-chosen buckets, so the tail (p99) is only as good as your bucket layout, and covering a range finely means emitting hundreds of `le` series per tag combo. Prometheus's adaptive answer — *native histograms* (exponential buckets, the closest thing to a T-Digest) — isn't emittable through Micrometer yet (experimental, protobuf-only as of late 2024). So the T-Digest PTF wins on tail accuracy **and** scales better: its sketch is bounded (~few KB/key) and mergeable, whereas at production volume a Micrometer percentile-histogram's `le`-bucket cardinality grows with the range you need to resolve. The built-in Flink `PERCENTILE` aggregate (exact, pure SQL) is *also* the wrong call at high volume — it retains every value in the window — so percentiles stay a **T-Digest sketch PTF** on purpose (see that file's header for the full rationale). This is a 3-Micrometer / 4-Flink split, not 4/3.

8.0 One-command showcase: Prometheus + Grafana on minikube

To *see* these meters instead of `curl`-ing `/metrics`, there's an optional, self-contained stack under k8s/monitoring/ — Prometheus + Grafana as minikube pods, with the datasource and a dashboard (all six produce/consume signals above) **auto-provisioned**, so it opens straight to a populated board with no login.

The pipeline stages run on your **host** via `./gradlew :app:run`, not in-cluster — so Prometheus scrapes back across the minikube→host bridge `host.minikube.internal`, one host port per stage (`enrich`→9410, `fulfill`→9411, `ship`→9412; edit k8s/monitoring/10-prometheus.yaml to change the mapping).

```
make metrics-up        # deploy, wait, port-forward Prometheus+Grafana, open
Grafana

# In separate terminals, run each stage on its mapped port (metrics on):
./gradlew :app:run -Dmetrics.prometheus.enabled=true -
Dmetrics.prometheus.port=9410 \
  -Disotope.consume.from=latest --args="enrich"     # fulfill→9411, ship→9412
# …then drive traffic with `place` so the meters move.
```

```
make metrics-down      # stop the port-forwards (pods stay up)
make metrics-delete    # tear the whole showcase down
```

Confirm scraping at Prometheus → Status → Targets (a target is `DOWN` until you start its stage — expected). Full runbook and troubleshooting: k8s/monitoring/README.md.